

POLITECNICO
MILANO 1863

Introduction

SENSOR SYSTEMS

Federica Villa
federica.villa@polimi.it

Prof. **Federica Villa**

phone 02-2399 **3490**

email federica.villa@polimi.it

office via Golgi 40 (building 24)

Ing. **Arianna De Vecchi**

email arianna.devecchi@polimi.it

office via Pascoli 70 (building 38)

FOR APPOINTMENTS: write an e-mail about 1 week in advance
and we will agree a day

Ing. **Ilaria Crupi**

email ilaria.crupi@polimi.it

office via Pascoli 70 (building 38)

Ing. **Christian Veronesi**

email christian.veronesi@polimi.it

office via Pascoli 70 (building 38)

FOR APPOINTMENTS: write an e-mail about 1 week in advance
and we will agree a day

24 hours theoretical classes

sensors and frontend circuits

26 hours hands-on classes (design laboratory)

using **STM32 microcontroller**

Day	hour	room	Activity
Monday	14:30 – 16:00	7.01	Lab Squad B
Wednesday	8:30 – 10:00	21.0.3	Theory
Friday	13:30 – 16:00	8.01	Lab Squad A

Theoretical classes can be attended in presence or through last year recordings.

Recordings of the labs will be available but use them wisely!

1. Sensors and front-end circuits

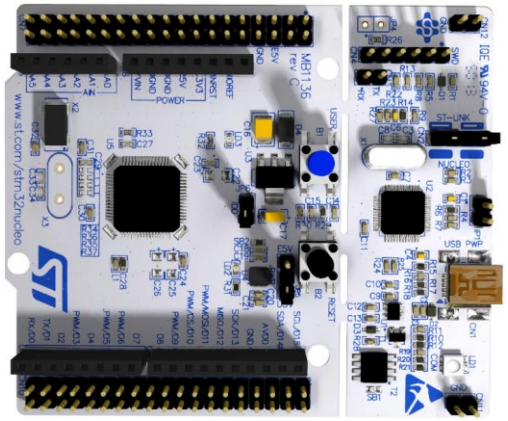
- Basics on sensors: definitions, classifications and characteristics.

- Working principle and front-end circuits of the most employed sensors:
  - light and imaging (photodiode, CCD and CMOS sensor);
  - temperature (RTD, thermistors, thermocouple, thermal diode, IR thermometer);
  - magnetic field (Hall sensors, magneto-resistances);
  - strain and forces (strain gauges, piezoelectric sensor);
  - displacement, distance and proximity (capacitive, inductive, acoustic, optical, magnetic, touch sensors);
  - acceleration and orientation (MEMS);
  - audio (dynamic and condenser microphones).



2. Microcontrollers for embedded sensors systems

- Introduction on ARM-Cortex M cores, STM32F4 microcontroller by STMicroelectronics and Nucleo development board.
- Basics on the main microcontrollers peripherals (GPIO, PWM, ADC, SPI, I2C, USART, etc.) and simple projects (pushbutton, blinking LED, sending strings to remote terminal, using communication interfaces, etc.).
- Sensors expansion board: hardware description and details of the sensors included in the board.
- Hands on activities to interface sensors (temperature, light, sound, rotation, accelerations) with the microcontroller using its peripherals and providing an output (sending data to pc or displaying on an LCD or generating a sound through a speaker).
- The main developed projects will include:
  - readout of a temperature smart sensor with I2C interface;
  - measure light intensity through a Light Dependent Resistor;
  - detect sounds with a microphone and generate souds with a speaker;
  - acquisition of the x, y, z accelerations with a tri-axial accelerometer;
  - communicate among boards with IR communication protocols



Friday

Squad A

Team A1

Team A2

Team A3

Team A4

Team A5

Team A6

Team A7

Monday

Squad B

Team B1

Team B2

Team B3

Team B4

Team B5

Team B6

Team B7

«If you want to go quick, go alone.

If you want to go far, go together!»

African proverbs

1) Written Test (75% of the final mark)

- 10 multiple choice questions (max 10 points)
  (sensors theory, microcontroller architecture and programming)
- 2 theoretical open questions (max 10 points)
- 2 laboratory open questions (max 10 points)

The written test will be done in the official exam date.

WeBeep “Supplementary materials”: 00 - Fac-simile - written test

2) Homework and short oral (25% of the final mark)

- 10 homework: each week of the course a short Team report should be provided
  - ✓ Homework assignments will not be graded individually.
  - ✓ The overall assessment will consider timely submission, completeness, and effort and engagement, including the ability to identify, document, and discuss problems encountered during the work.
  - ✓ **Keep your reports concise:** you are not expected to spend hours writing detailed reports. Briefly summarize what you did, what you learned, and any difficulties you encountered.

- Short oral will consist in a discussion of the homework (you are expected to brig your laptop with all the projects done during the course)

The short oral will be about one week after the written test, with possibility to agree for a different date with the professor.

WeBeep “Supplementary materials”: 01 – Homework report template

- The **labs** will be organized into three parts:
    1. a brief review of the homeworks, highlighting the main points for their resolution;
    2. a theoretical section about microcontroller architecture, peripherals and programming
    3. a subsequent practical one, focusing on a project related to the lab topic. At the end of the lab session, the teacher will provide a possible solution.

- Two SS boards will be provided to each Team.

  Please ensure you **bring your laptop and the SS board** to every lab session.

- **Attendance** is strongly recommended!

Teams organization

- You will **choose your group members**.

  We will provide you with an Excel file (**Teams_organization**) where you can create your Team. The link to the file will be added to the materials tab in Webeep and provided via e-mail, so be sure to enrol on the Webeep course as soon as possible.

- If you cannot create a Team on your own, write your name in an empty/partially filled group, and we will create Teams for these remaining people.

- Team members: 4-5 (preferably from different degree programs)

Homework

- After each lab session, a **homework** assignment will be given.
- The due date for each lab assignment is **24 hours prior to the start of the subsequent lab session** for the respective team.
- Each homework task involves completing a **project** and preparing a **report** that summarizes the crucial aspects and primary challenges encountered.
- The report must be uploaded in the **Assignments** tab in the WeBeep page of the course (one per Team).
- One assignment per homework will be created.
- The **oral exam will be a discussion of the homework**, so it is important all projects work correctly, even if you are not able to complete them for the due date.
- Homework will contribute to the final mark, as explained in the Assessment modalities.

- Microcontrollers
  Franco Zappa – Esculapio

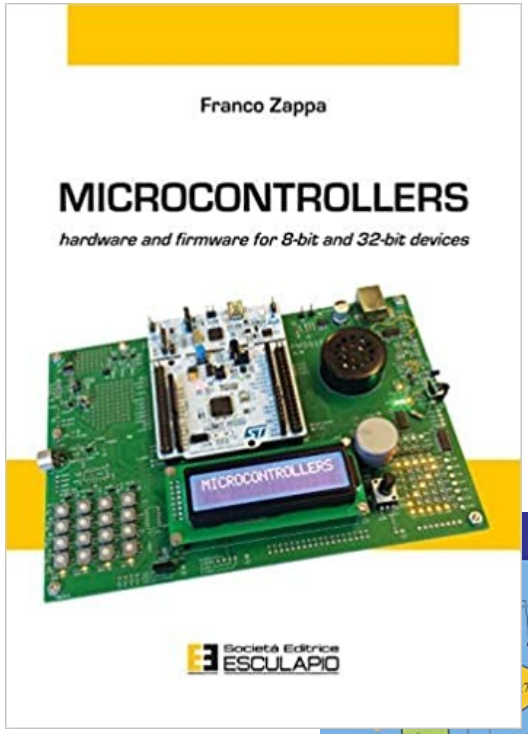
- Electronic Systems
  Franco Zappa – Esculapio

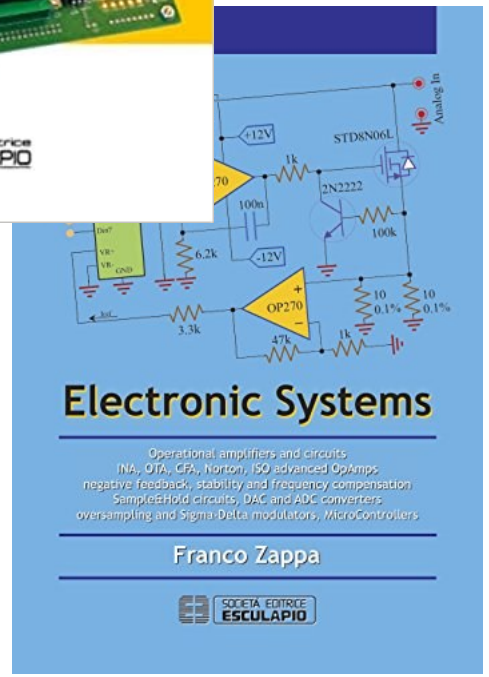
- Handbook of Modern Sensors - Physics, Designs, and Applications
  Jacob Fraden - Springer

- Electronics Sensors for the Evil Genius - 54 electrifying projects
  Tom Petruzzellis - TAB Electronics

- Make: Getting Started with Sensors - Measure the Word with Electronics, Arduino, and Raspberry Pi
  Kimmo Karvinen & Tero Karvinen - makezine.com

- Measurement, Instrumentation, and Sensors Handbook
  John G. Webster, Halit Eren - CRC press





A practical example

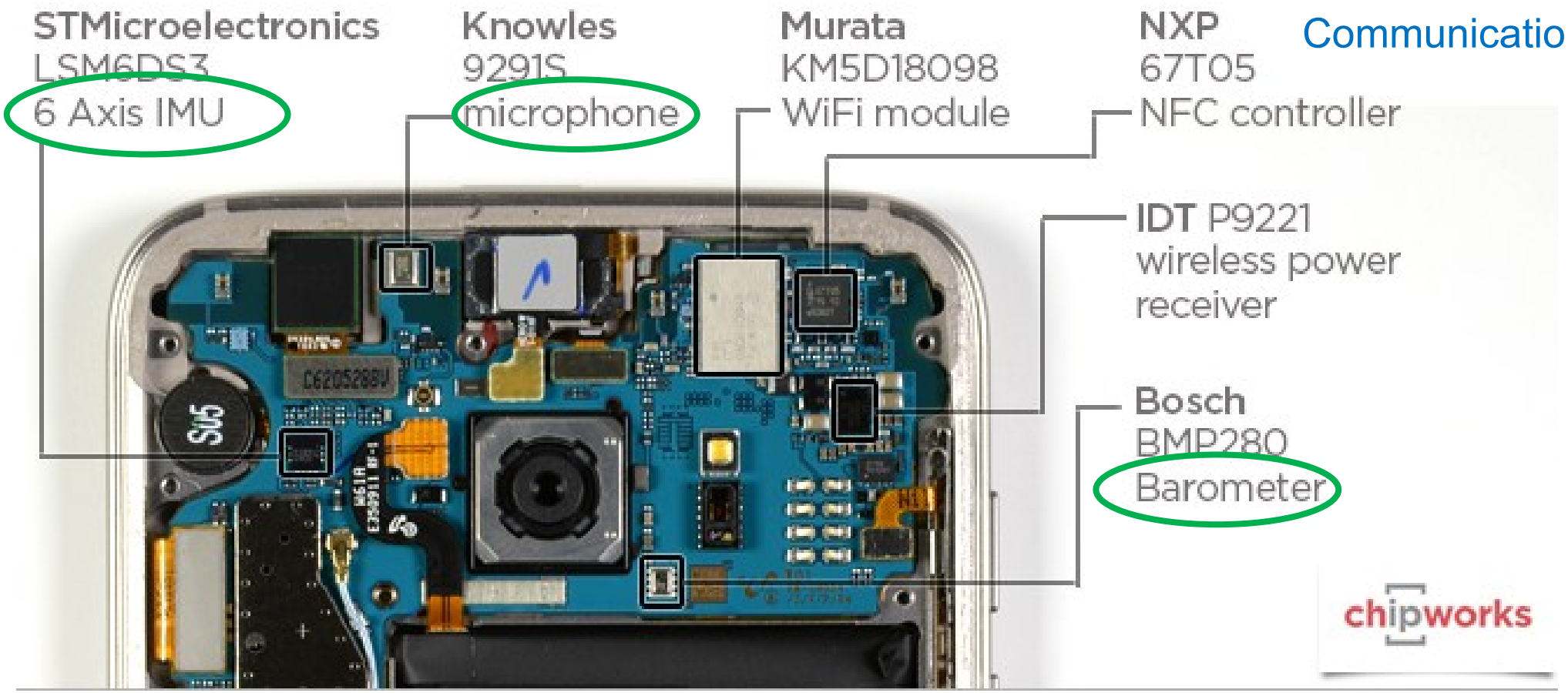


https://www.techinsights.com/

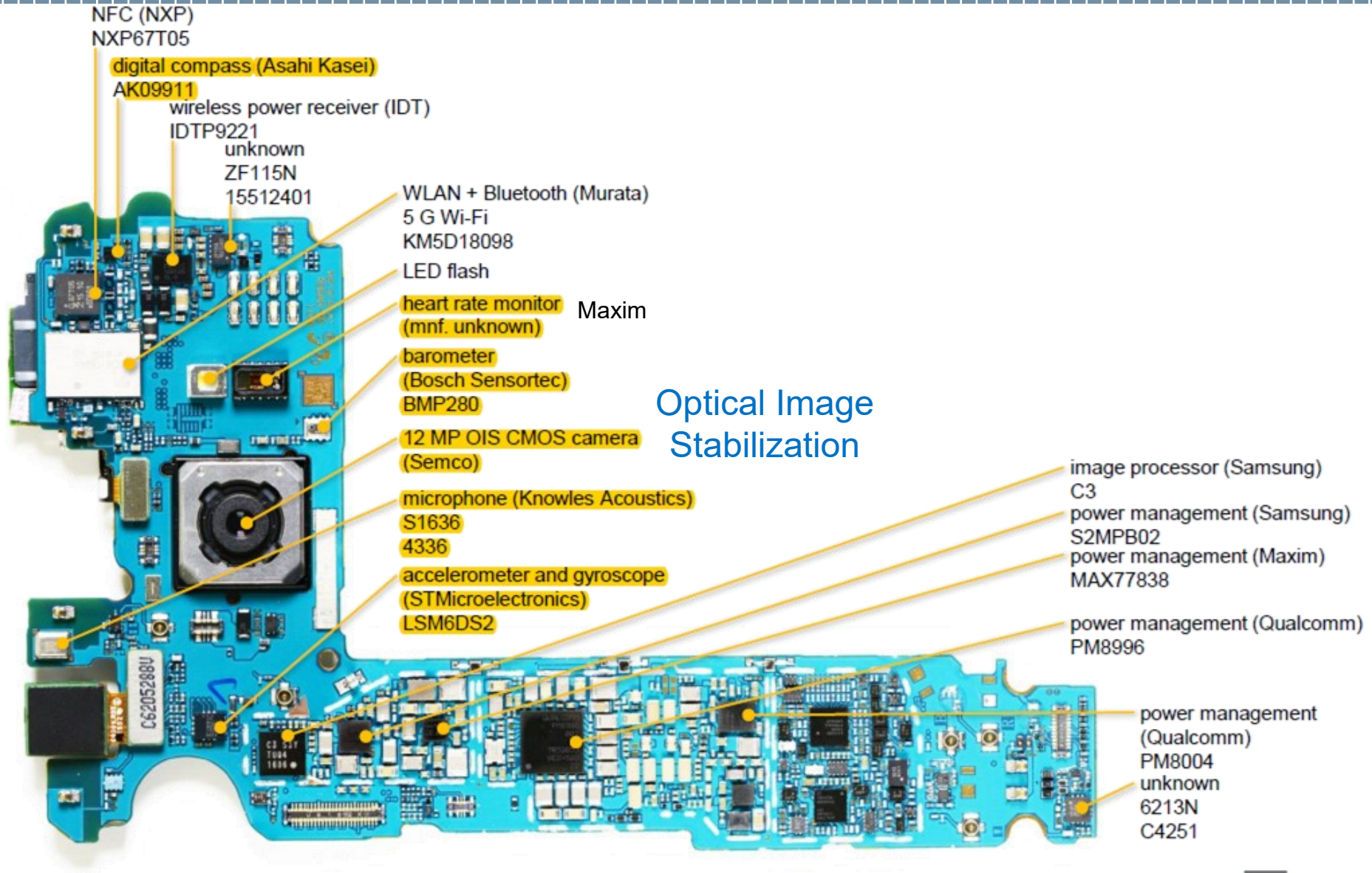
WeBeep “Supplementary materials”: 02 - Galaxy s7 edge

Inertial Measurement Unit

Near Field Communication



PCB – Battery Side



chipworks

Galaxy S7 edge sensors

Sensor	Manufacturer	Code
CMOS image sensor 12Mpx (rear camera)	Sony	IMX260
CMOS image sensor 5Mpx (front camera)	Samsung	S5K4E6
Microphone	Knowles	SPH0641LM4H-1
Pulse Oximeter	Maxim	MAX30100
Proximity sensor	AMS	TMD490X
Barometer (pressure sensor)	Bosch	BMP280
Gyroscope	STMicroelectronics	L2G2IS
6 axis IMU (Inertial Measurement Unit)	STMicroelectronics	LSM6DS3
Electronic compass	AKM	AK09911
Touch Screen Controller	Samsung	S6SA552X

Resolution **12 Mpixel**

Pitch **1.44 µm**

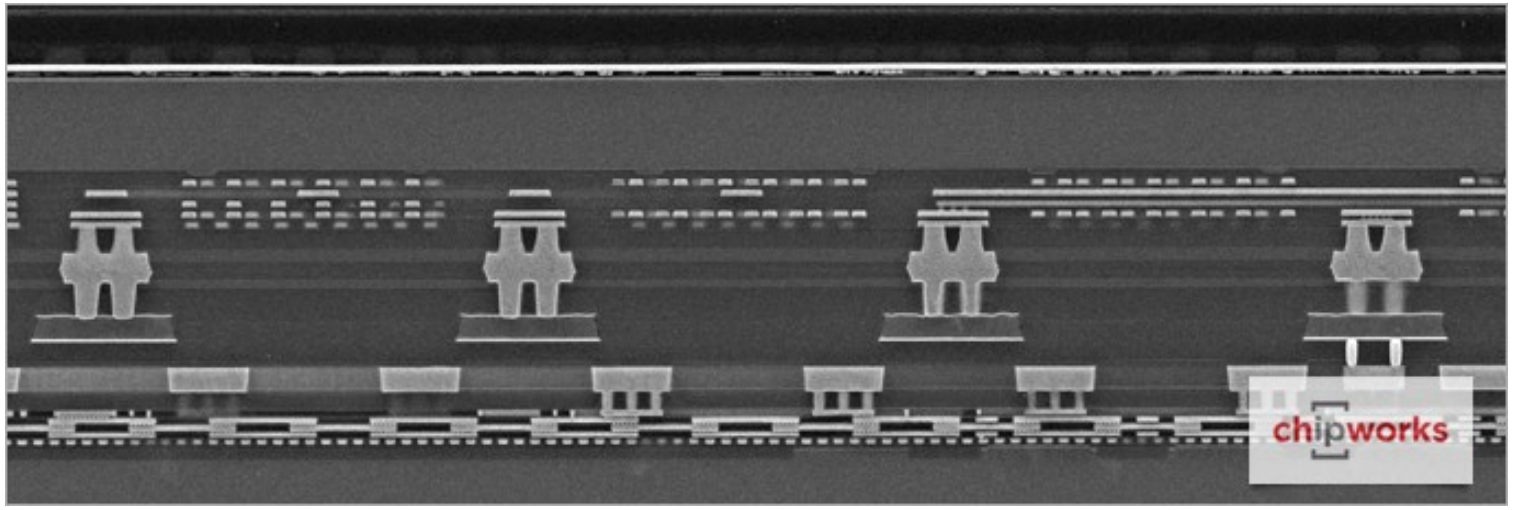
Sony back-illuminated (Exmor R) CMOS image sensor

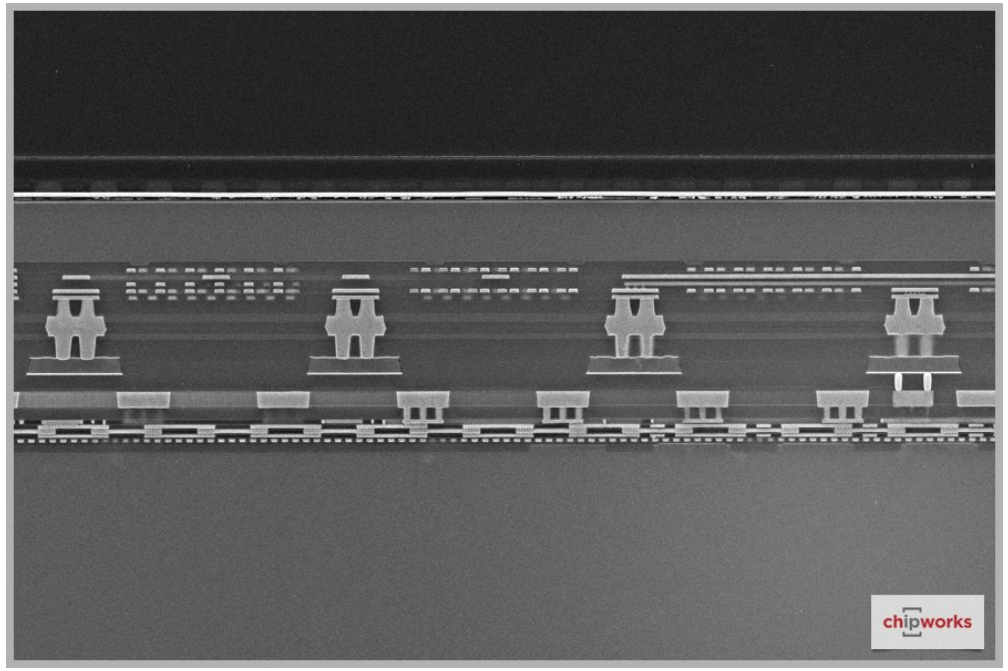
implemented with a 3D-stacked hybrid-bonding (CIS + ISP) solution:
CIS = 5 metal CMOS Image Sensor
ISP = 7 metal Image Signal Processor

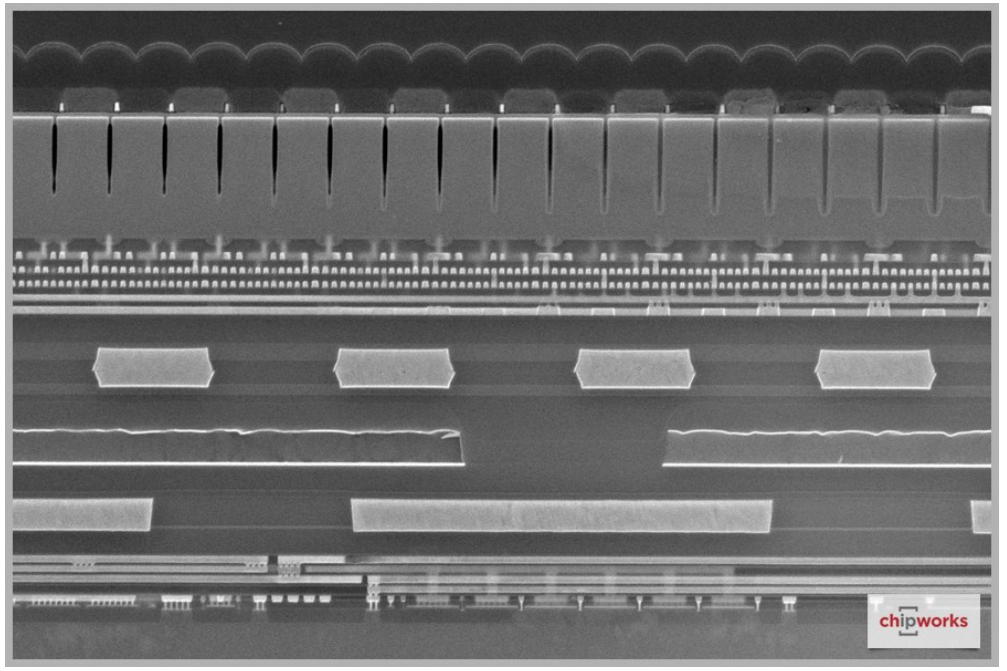
Dual pixel phase detection **autofocus** (PDAF)

(100% of the active pixel array committed to both sensing and AF system)

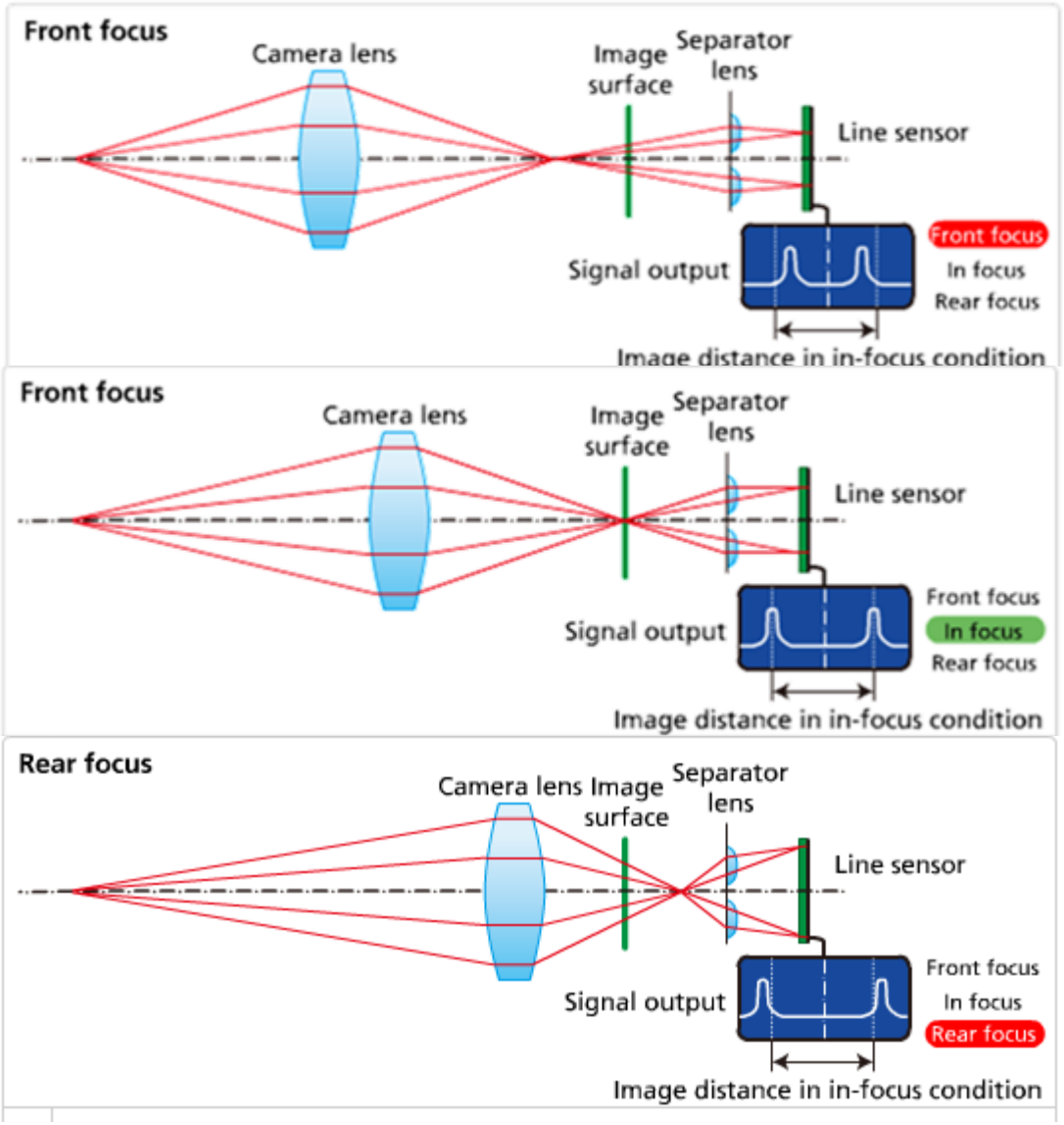
CMOS image sensor

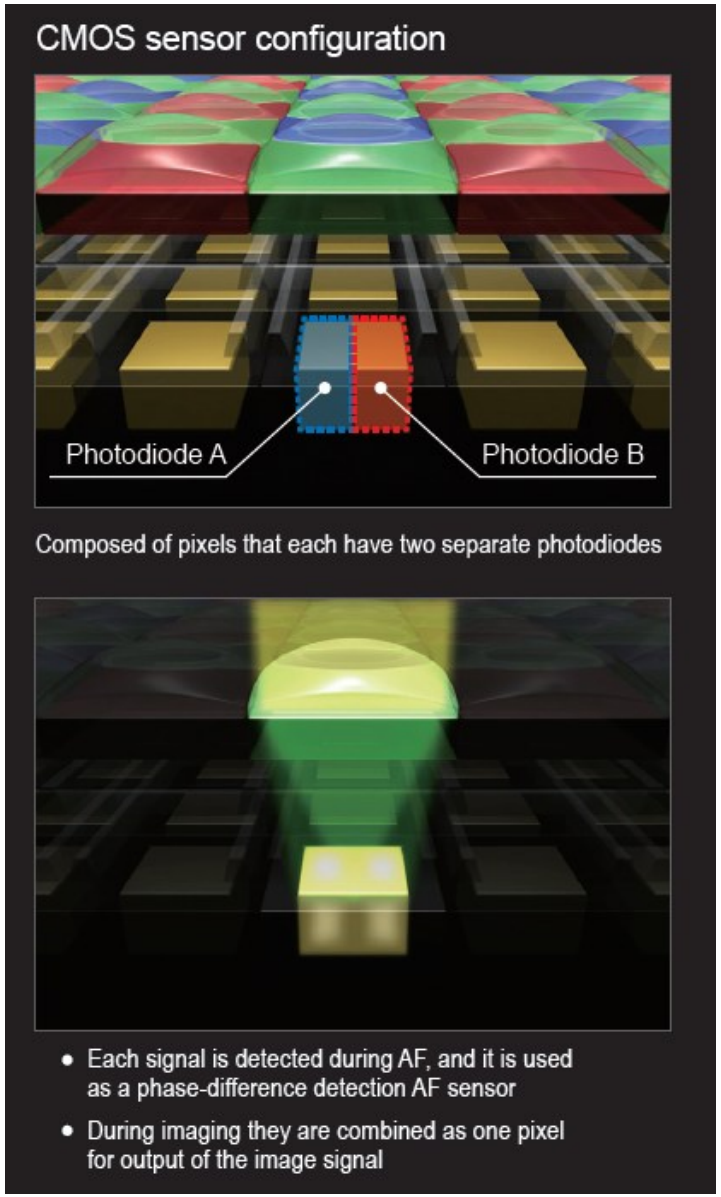






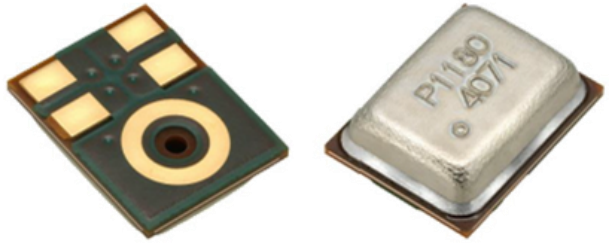
Dual pixel phase detection autofocus





Digital Zero-Height SiSonic™ Microphone With Multiple Performance Modes

Knowles

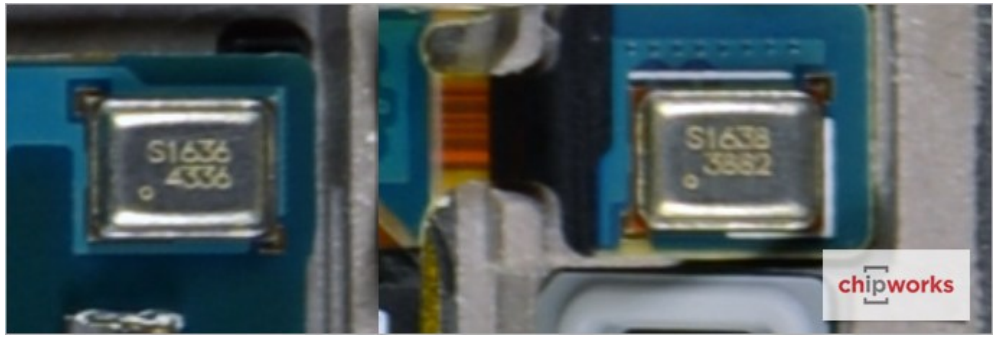


Pulse Density Modulation

The SPH0641LM4H-1 is a miniature, high-performance, low power, bottom port silicon digital microphone with a single bit PDM output. Using Knowles' proven high performance SiSonic™ MEMS technology, the SPH0641LM4H-1 consists of an acoustic sensor, a low noise input buffer, and a sigma-delta modulator. These devices are suitable for applications such as cellphones, smart phones, laptop computers, sensors, digital still cameras, portable music recorders, and other portable electronic devices where excellent wideband audio performance and RF immunity are required. In addition, the SPH0641LM4H-1 offers multiple performance modes

Features:

- High SNR of 64dB
- Low Current Consumption of 230uA in Low-Power Mode
- Flat Frequency Response
- RF Shielded
- Zero-Height Mic™
- Supports Dual Multiplexed Channels
- Standard SMD Reflow
- Omnidirectional
- Multiple performance modes (Sleep, Low-Power, Standard Performance)
- Sensitivity Matching
- Small Size



Alpha Character A:

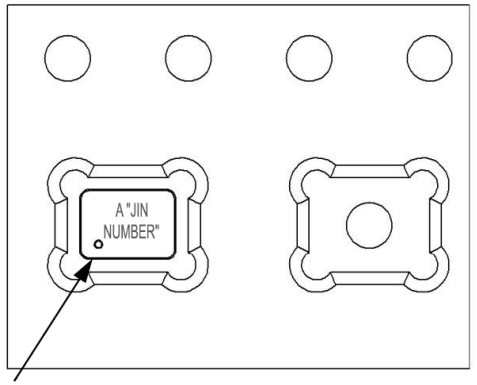
"S": Knowles SiSonic™ Production

"E": Knowles Engineering Samples

"P": Knowles Prototype Samples

"JIN NUMBER":

Unique Job Identification Number for product traceability



Pin #	Pin Name	Type	Description
1	DATA	Digital O	PDM Output
2	SELECT	Digital I	Lo/Hi (L/R) Select This pin is internally pulled low but should not be left floating.
3	GROUND	Power	Ground
4	CLOCK	Digital I	Clock Input
5	V_{DD}	Power	Power Supply

MAX30100

Pulse Oximeter and Heart-Rate Sensor IC for Wearable Health

maxim integrated

General Description

The MAX30100 is an integrated pulse oximetry and heart-rate monitor sensor solution. It combines two LEDs, a photodetector, optimized optics, and low-noise analog signal processing to detect pulse oximetry and heart-rate signals.

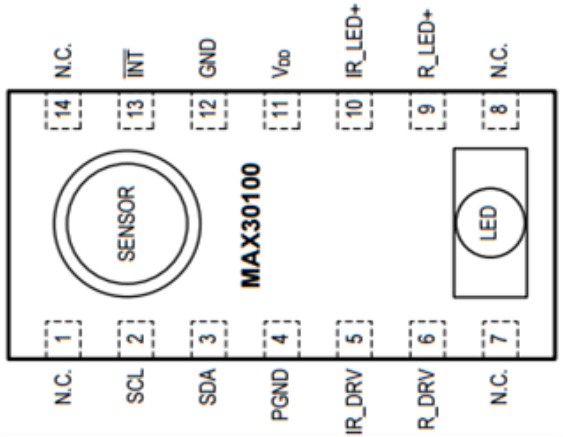
The MAX30100 operates from 1.8V and 3.3V power supplies and can be powered down through software with negligible standby current, permitting the power supply to remain connected at all times.

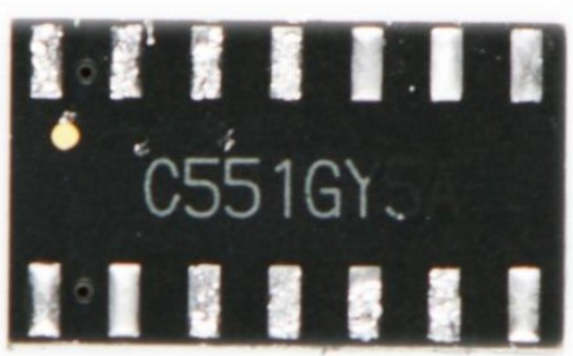
Applications

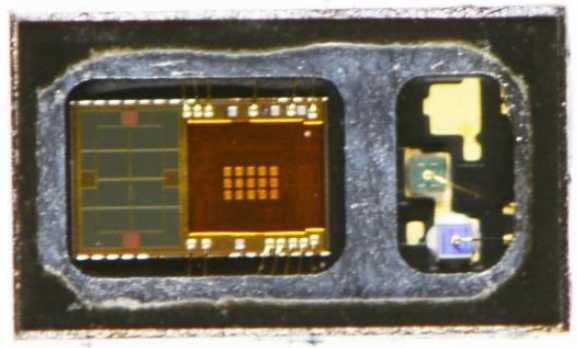
- Wearable Devices
- Fitness Assistant Devices
- Medical Monitoring Devices

Benefits and Features

- Complete Pulse Oximeter and Heart-Rate Sensor Solution Simplifies Design
  - Integrated LEDs, Photo Sensor, and High-Performance Analog Front -End
  - Tiny 5.6mm x 2.8mm x 1.2mm 14-Pin Optically Enhanced System-in-Package
- Ultra-Low-Power Operation Increases Battery Life for Wearable Devices
  - Programmable Sample Rate and LED Current for Power Savings
  - Ultra-Low Shutdown Current (0.7µA, typ)
- Advanced Functionality Improves Measurement Performance
  - High SNR Provides Robust Motion Artifact Resilience
  - Integrated Ambient Light Cancellation
  - High Sample Rate Capability
  - Fast Data Output Capability

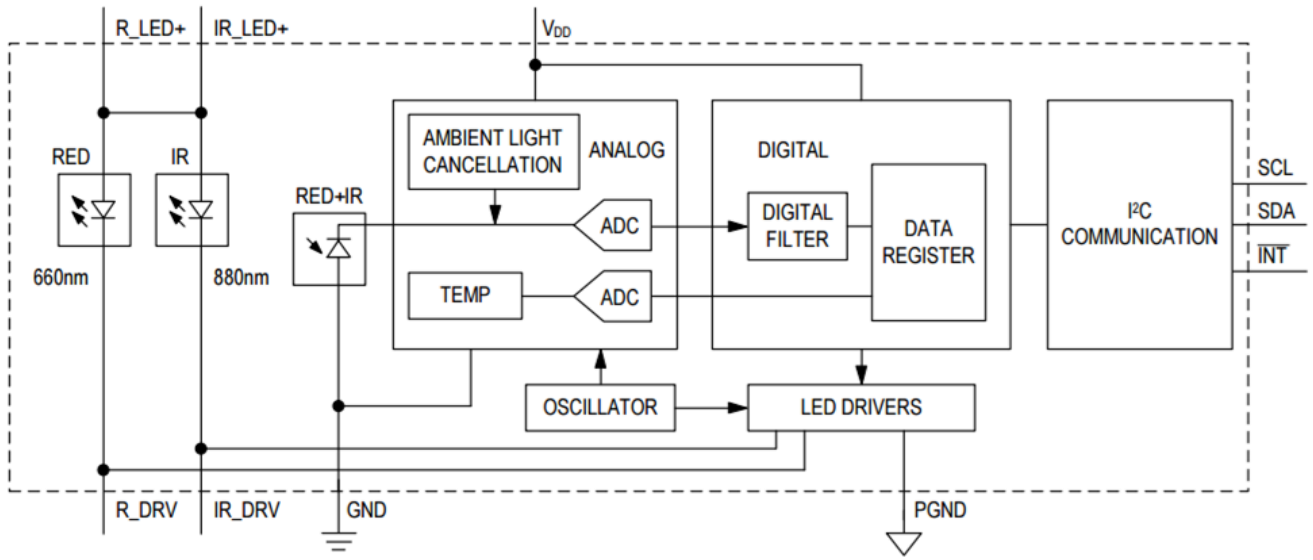


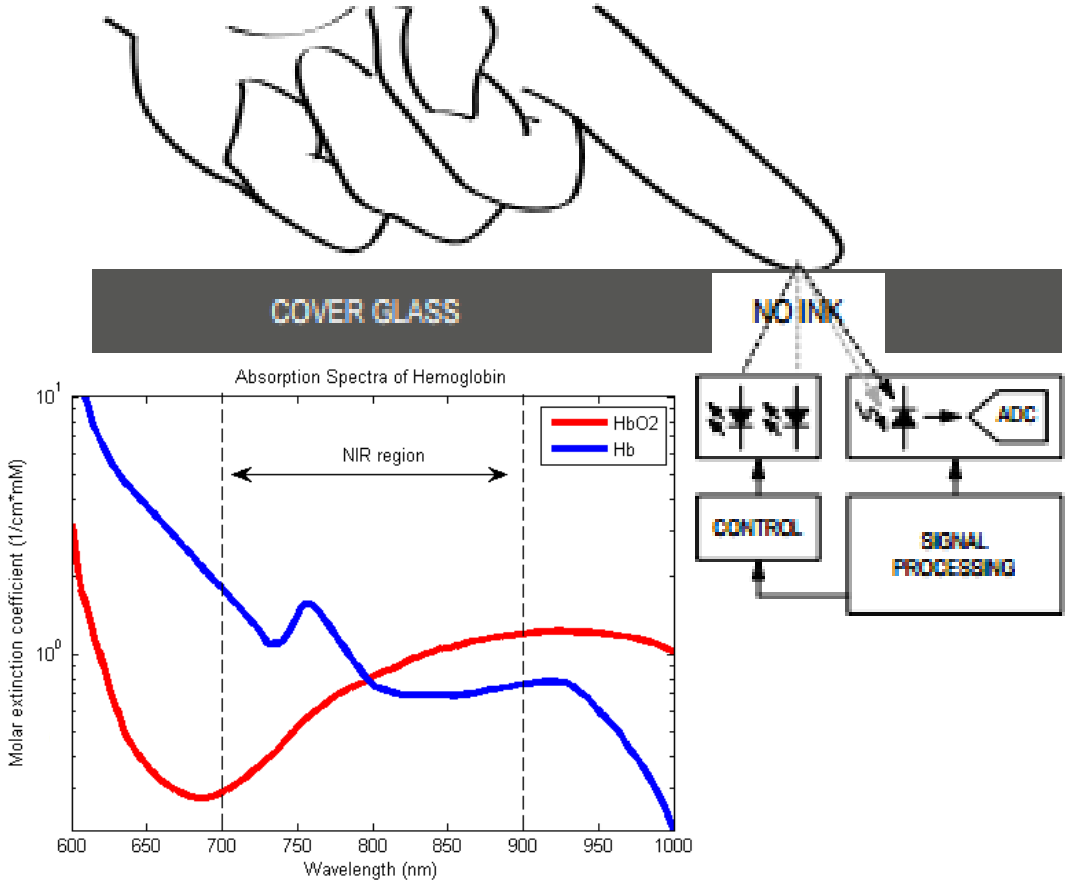


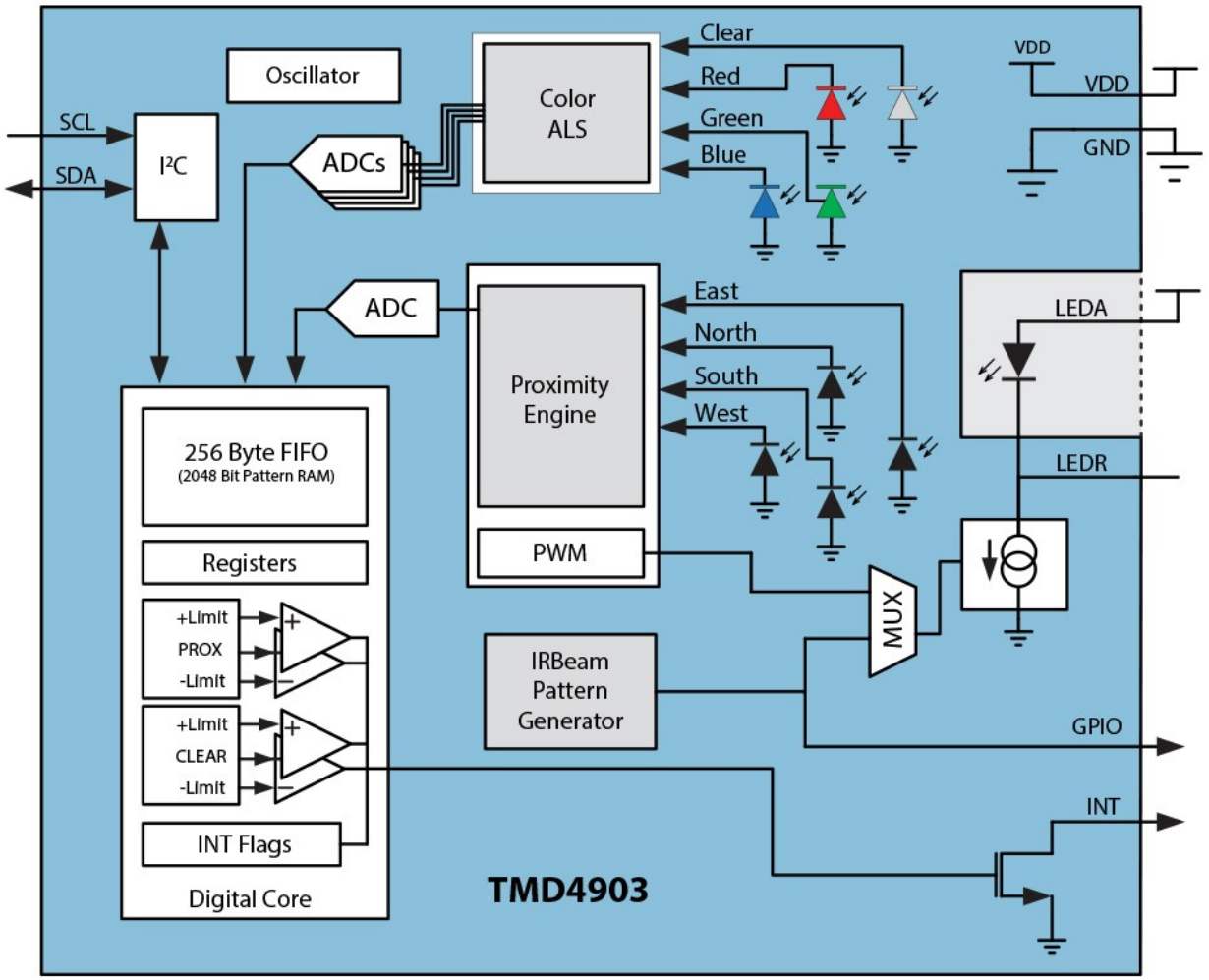


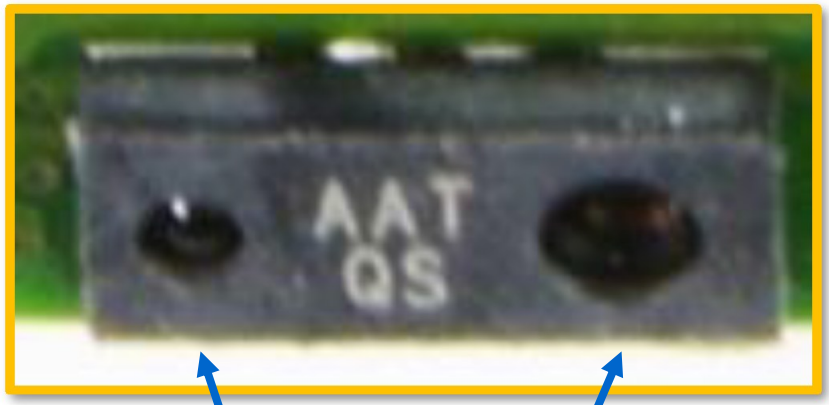
2 LEDs (RED and IR)

1 photodiode









light/proximity sensor (AMS)
TMD490X

Optical windows for LED and photodiodes

BMP280
Digital, barometric pressure sensor

Bosch Sensortec

BOSCH
Invented for life

General description

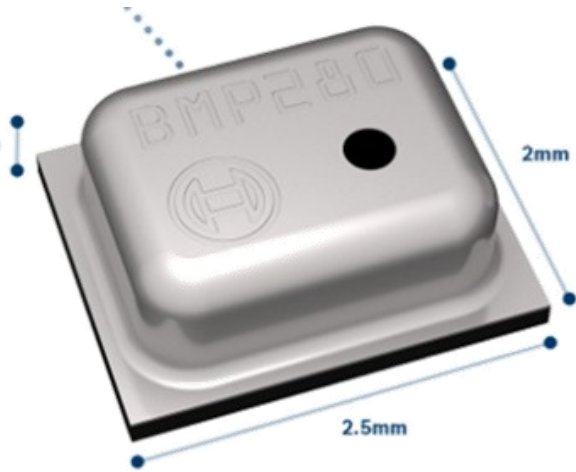
BMP280 is an absolute barometric pressure sensor especially designed for mobile applications. The sensor module is housed in an extremely compact package. Its small dimensions and its low power consumption allow for the implementation in battery driven devices such as mobile phones, GPS modules or watches.

As its predecessor BMP180, the BMP280 is based on Bosch's proven piezo-resistive pressure sensor technology featuring high accuracy and linearity as well as long term stability and high EMC robustness.

Numerous device operation options offer highest flexibility to optimize the device regarding power consumption, resolution and filter performance. A tested set of default settings for example use case is provided to the developer in order to make design-in as easy as possible.

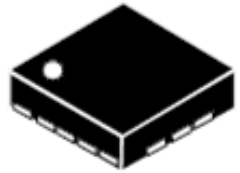
Key parameters

• Pressure range	300 ... 1100 hPa (equiv. to +9000...-500 m above/below sea level)
• Package	8-pin LGA metal-lid Footprint : 2.0 × 2.5 mm², height: 0.95 mm
• Relative accuracy (950 ... 1050hPa @25°C)	±0.12 hPa, equiv. to ±1 m
• Absolute accuracy (950 ...1050 hPa, 0 ...+40 °C)	typ. ±1 hPa
• Temperature coefficient offset (25 ... 40°C @900hPa)	1.5 Pa/K, equiv. to 12.6 cm/K
• Digital interfaces	I²C (up to 3.4 MHz) SPI (3 and 4 wire, up to 10 MHz)
• Current consumption	2.7µA @ 1 Hz sampling rate
• Temperature range	-40 ... +85 °C



L2G2IS

MEMS motion sensor:
ultra-compact two-axis gyroscope for optical image stabilization



LGA-16 (2.3x2.3x0.7 mm)

Description

The L2G2IS is a two-axis MEMS gyroscope for optical image stabilization applications.

It includes a sensing element and an IC interface capable of providing the measured angular rate to the application through an SPI digital interface.

The unique sensing element is manufactured using a dedicated micromachining process developed by STMicroelectronics to produce inertial sensors and actuators on silicon wafers.

The IC interface is manufactured using a CMOS process that allows a high level of integration to design a dedicated circuit which is trimmed to better match the characteristics of the sensing element.

The L2G2IS is available in a plastic land grid array (LGA) package and can operate over a temperature range of -40 °C to +85 °C.

degree per second

Features

- ±100 dps / ±200 dps full-scale range
- Wide supply voltage range: 1.71 V to 3.6 V
- Low-voltage compatible IOs
- 3- and 4-wire SPI digital interface
- Embedded temperature sensor
- Embedded self-test
- Integrated low- and high-pass filters with user-selectable bandwidth
- Power-down and sleep modes for smart power saving
- ECOPACK®, RoHS and "Green" compliant

Applications

- Optical image stabilization

12 MP OIS CMOS camera (rear)

Optical Image Stabilization

OIS driver (Renesas)
RAY00

gyroscope (STMicroelectronics)
L2G2IS

serial flash memory (Winbond)
W25Q32FW







life.augmented

LSM6DS3

iNEMO inertial module:
always-on 3D accelerometer and 3D gyroscope

Description

The LSM6DS3 is a system-in-package featuring a 3D digital accelerometer and a 3D digital gyroscope performing at 1.25 mA (up to 1.6 kHz ODR) in high-performance mode and enabling always-on low-power features for an optimal motion experience for the consumer.

The LSM6DS3 supports main OS requirements, offering real, virtual and batch sensors with 8 kbyte for dynamic data batching.

ST's family of MEMS sensor modules leverages the robust and mature manufacturing processes already used for the production of micromachined accelerometers and gyroscopes.

The various sensing elements are manufactured using specialized micromachining processes, while the IC interfaces are developed using CMOS technology that allows the design of a dedicated circuit which is trimmed to better match the characteristics of the sensing element.

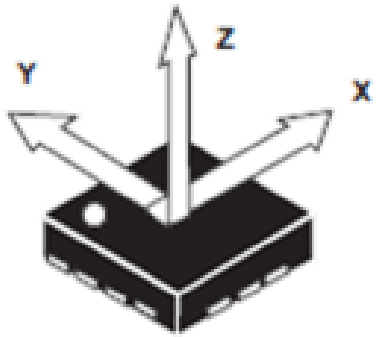
The LSM6DS3 has a full-scale acceleration range of ±2/±4/±8/±16 *g* and an angular rate range of ±125/±245/±500/±1000/±2000 dps.

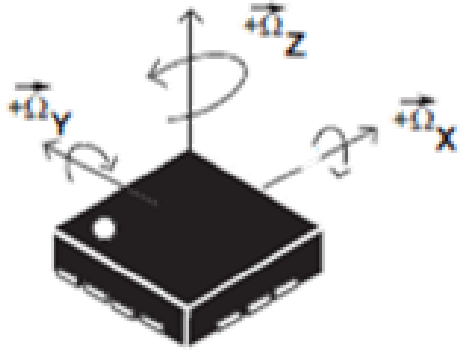
High robustness to mechanical shock makes the LSM6DS3 the preferred choice of system designers for the creation and manufacturing of reliable products.

The LSM6DS3 is available in a plastic land grid array (LGA) package.



LGA-14L
(2.5 x 3 x 0.83 mm) typ.





Features

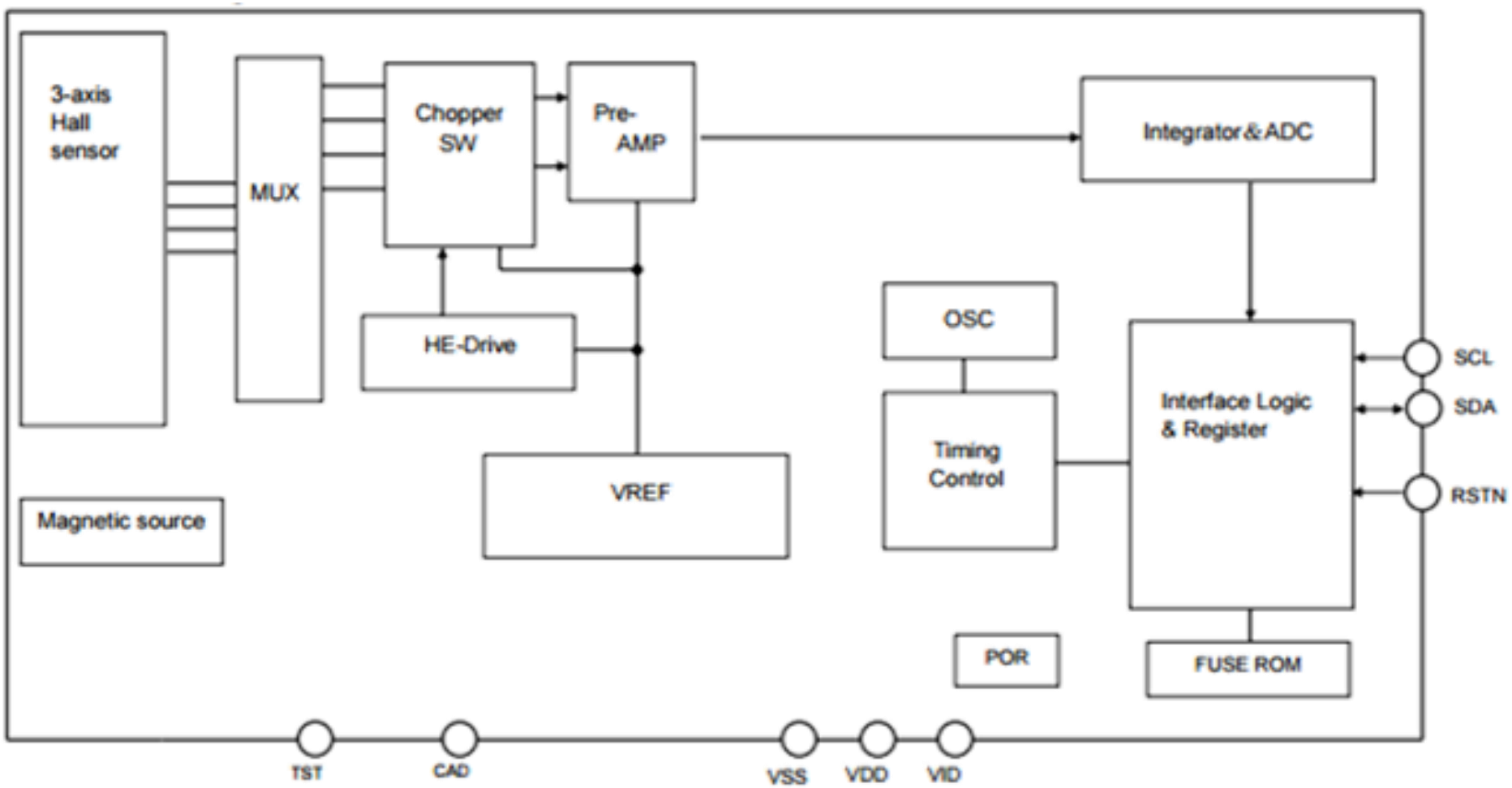
- Power consumption: 0.9 mA in combo normal mode and 1.25 mA in combo high-performance mode up to 1.6 kHz
- "Always-on" experience with low power consumption for both accelerometer and gyroscope
- Smart FIFO up to 8 kbyte based on features set
- Compliant with Android K and L
- Hard, soft ironing for external magnetic sensor corrections
- ±2/±4/±8/±16 *g* full scale
- ±125/±245/±500/±1000/±2000 dps full scale
- Analog supply voltage: 1.71 V to 3.6 V
- Independent IOs supply (1.62 V)
- Compact footprint, 2.5 mm x 3 mm x 0.83 mm
- SPI/I^{2}C serial interface with main processor data synchronization feature
- Embedded temperature sensor
- ECOPACK®, RoHS and "Green" compliant

Applications

- Pedometer, step detector and step counter
- Significant motion and tilt functions
- Indoor navigation
- Tap and double-tap detection
- IoT and connected devices
- Intelligent power saving for handheld devices
- Vibration monitoring and compensation
- Free-fall detection
- 6D orientation detection

AsahiKASEI [AK09911]

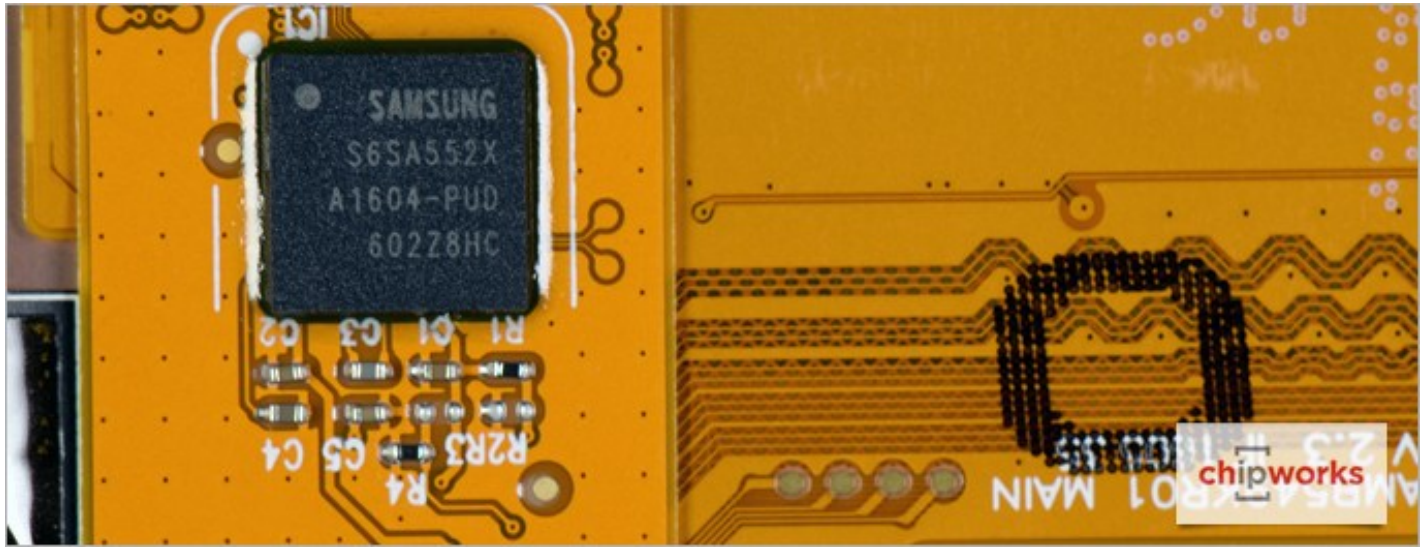
4. Circuit Configration





Touch Screen panel and controller

Samsung S6SA552X – Capacitive touch screen



SPECIFICATIONS
TOUCH CONTROLLER › S6SA551

Scan/Report rate	160Hz/120Hz
Channel	52
Package	FBGA (Fine Pitch Ball Grid Array)
VDD max	1.65V

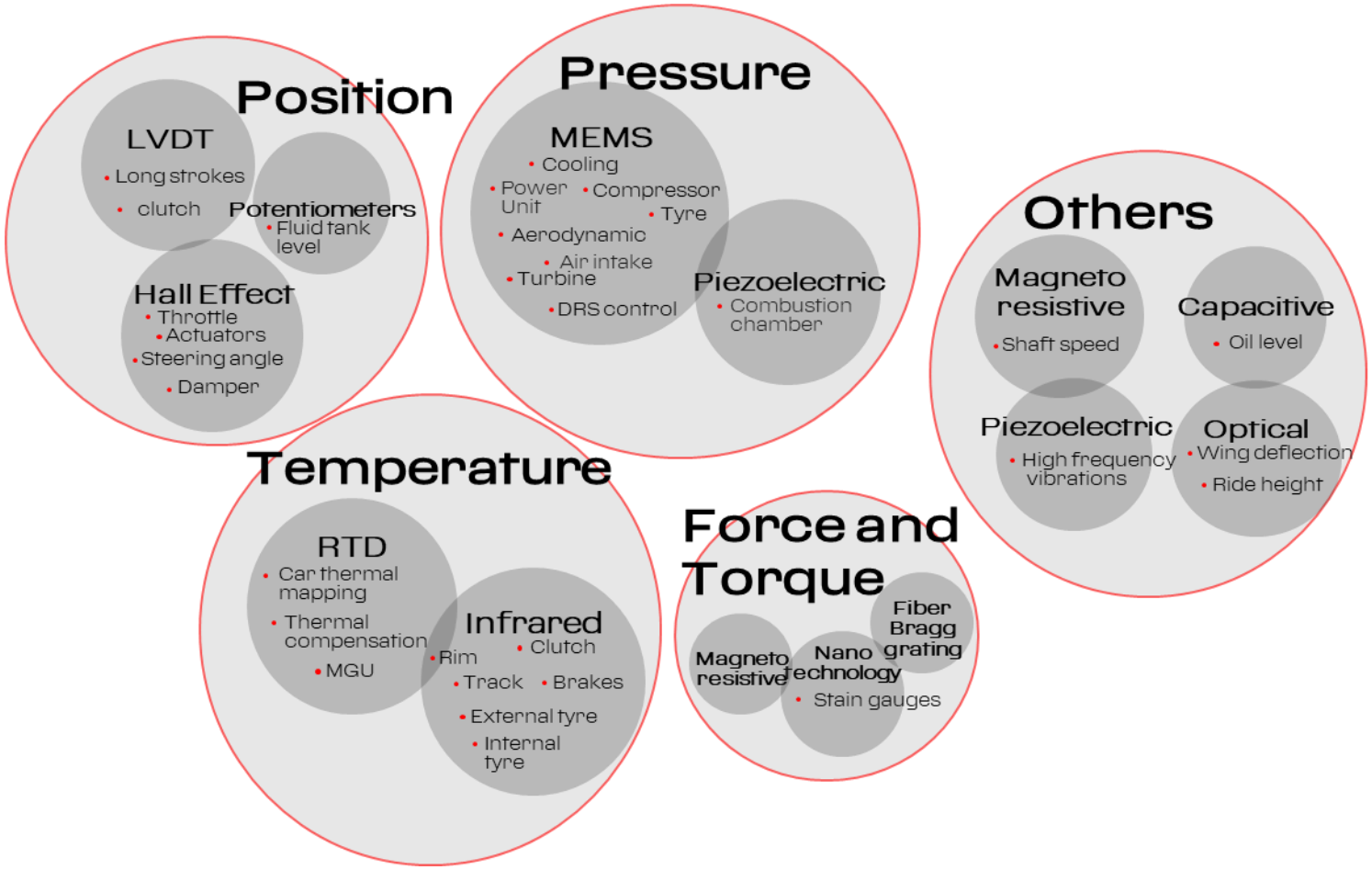


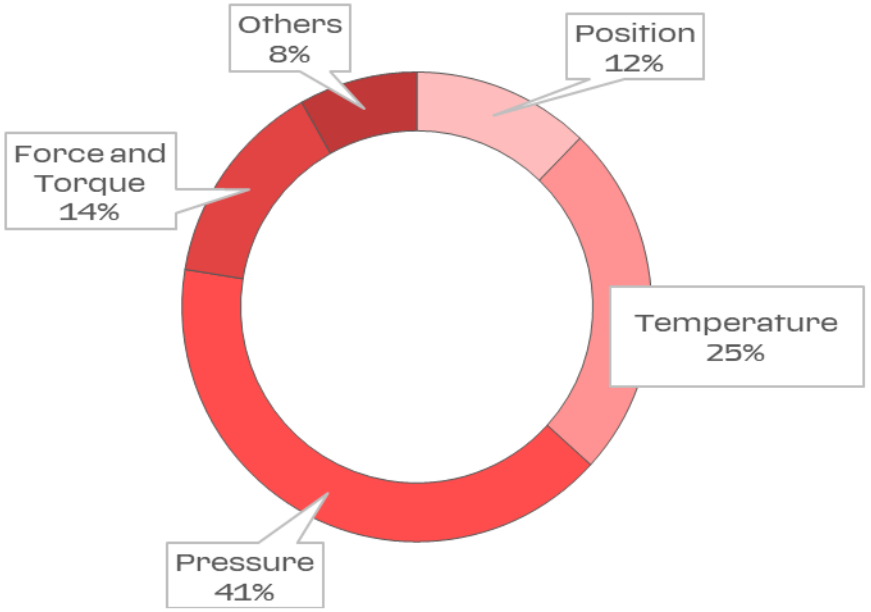
Sensors

300	1200	14	0.38%
TOTAL SENSORS FITTED	MEASUREMENTS GENERATED	TECHNOLOGY TYPES	SENSOR TO CAR TOTAL MASS

55	60	185	1943
PU CONTROL SENSORS	CHASSIS CONTROL SENSORS	“MONITORING” SENSORS	METERS OF CABLING FOR SENSOR DATA ACQUISITION

Sensors technologies





EssilorLuxottica

Smart Vision



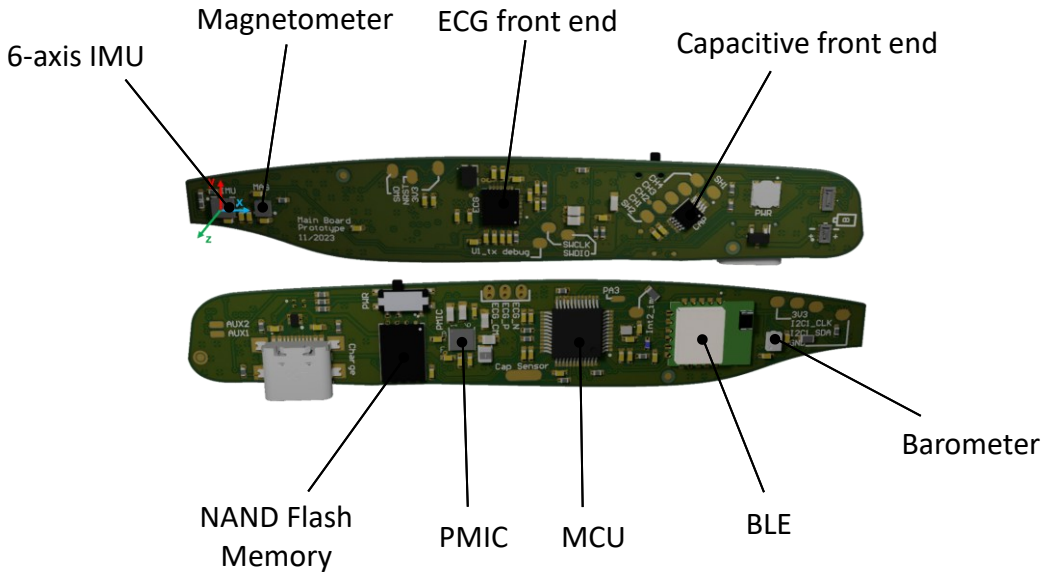
Smart Health

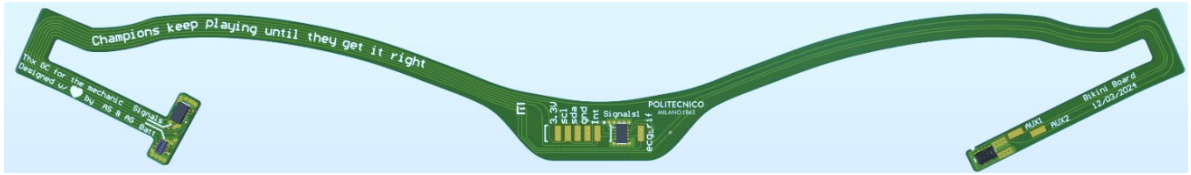
Smart Light protection

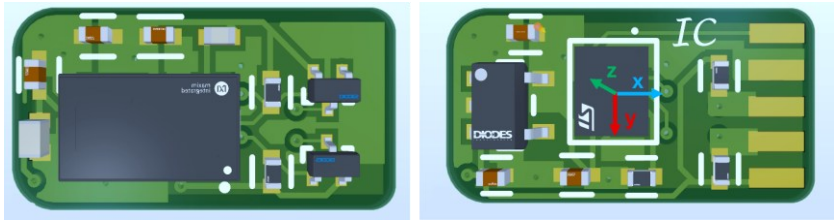
Smart Communication

Smart eyewear for biomarkers monitoring









MCU: STM32U575 - ARM Cortex-M33, 2MB Flash, 784KB SRAM